

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE:ITA0604

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:100

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I Jeevan Raaja S (192421005) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of
the student:
Total
Marks
Awarded***

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	20%	15%	15%	
Q1						
Q2						
Q3						

Annexure A

Short Answer Test

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

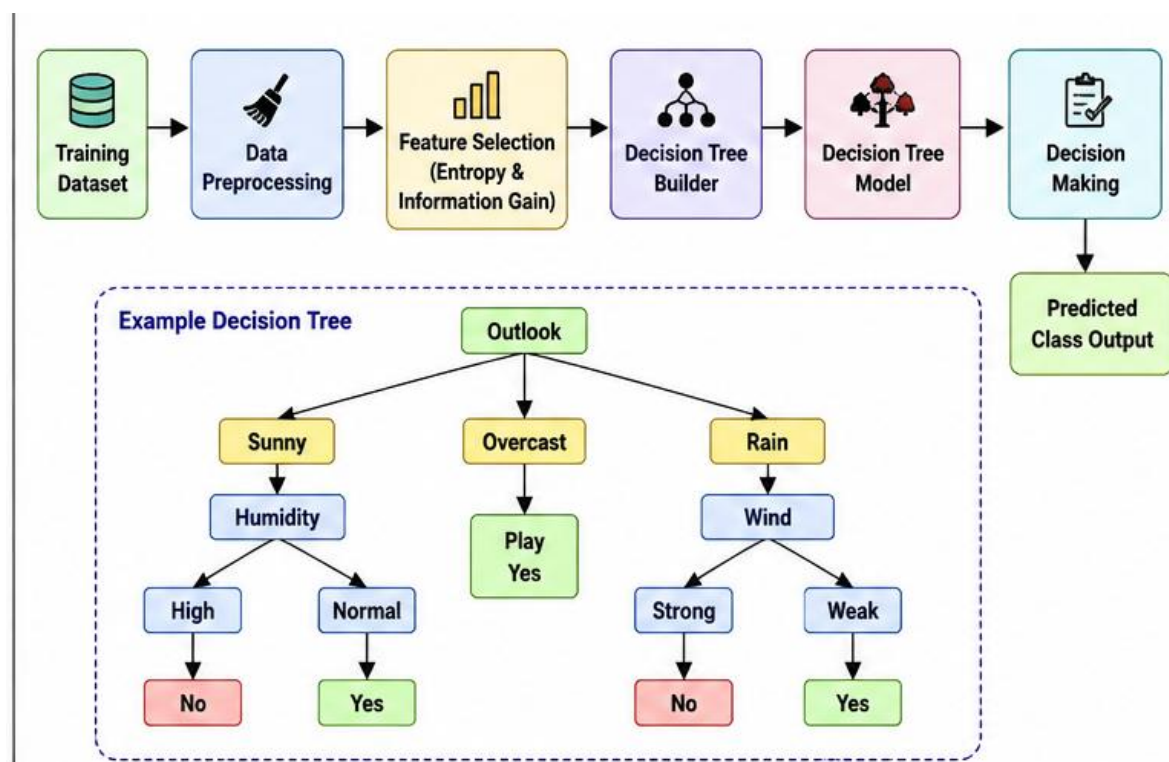
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models.

Answer

A Decision Tree is a supervised machine learning algorithm used for both classification and regression tasks. It represents decisions in the form of a tree structure, where each internal node represents a test on a feature, each branch represents the outcome of the test, and each leaf node represents the final prediction.



Feature Selection

The decision tree chooses the best feature for splitting the data using **Entropy** and **Information Gain**.

Entropy measures the impurity or randomness in the dataset.

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

- Entropy = 0 → Pure dataset

- Higher entropy → More uncertainty

Information Gain (IG) measures the reduction in entropy after splitting the data.

$$IG = Entropy(Parent) - Entropy(Children)$$

The feature with the **highest Information Gain** is selected as the root node.

Tree Construction Example

Outlook Humidity Play

Sunny	High	No
Sunny	Normal	Yes
Rain	High	No
Overcast	High	Yes

The algorithm calculates the information gain for Outlook and Humidity. If Outlook has the highest information gain, it becomes the root node. The process continues recursively until all records are classified.

Pruning

Pruning removes unnecessary branches that do not improve prediction accuracy.

- **Pre-Pruning:** Stops tree growth early by limiting tree depth or minimum samples.
- **Post-Pruning:** Builds the complete tree first and then removes unimportant branches.

Controlling Overfitting

Overfitting occurs when the model memorizes the training data instead of learning general patterns.

It can be controlled by:

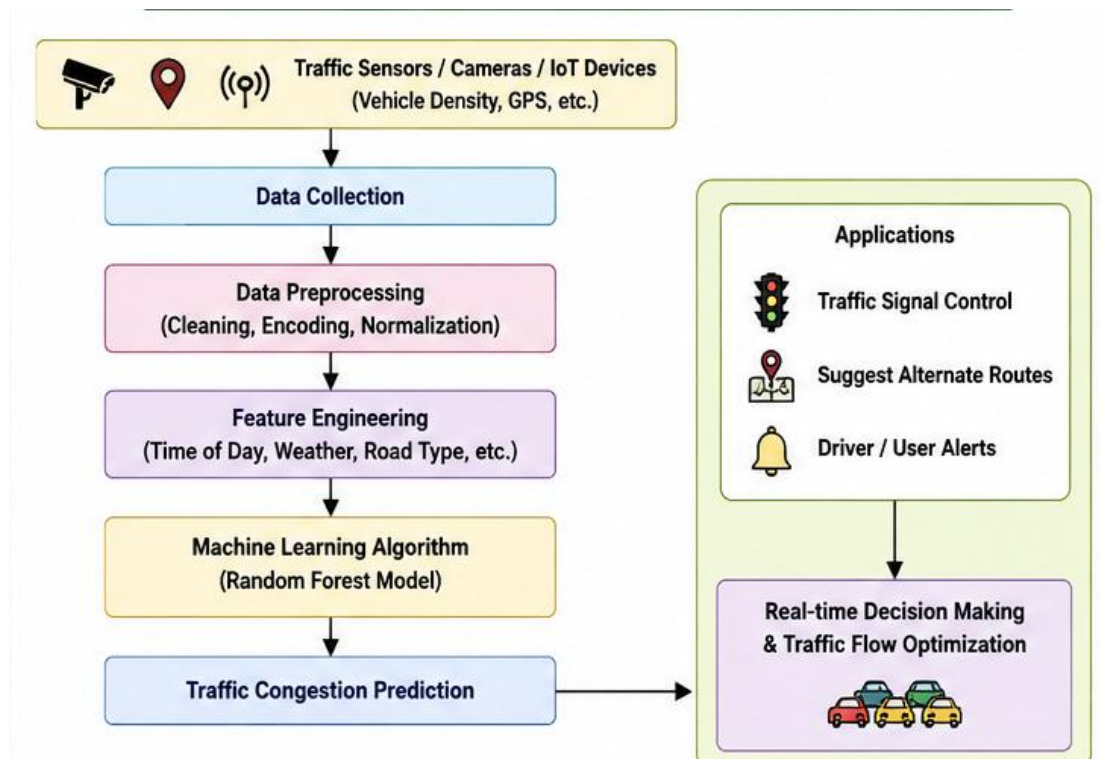
- Limiting maximum tree depth.
- Setting a minimum number of samples required for splitting.
- Using pre-pruning and post-pruning techniques.
- Validating the model using cross-validation.

Conclusion: Decision trees are simple, interpretable, and efficient models. Proper feature selection and pruning improve accuracy while reducing overfitting.

2. **Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making.**

Answer

A smart traffic management system uses machine learning to predict congestion and assist traffic authorities in reducing delays and improving road safety.



Input Features

- Vehicle density
- Time of day
- Weather conditions
- Road type
- Traffic speed
- Accident reports

Algorithm Selection

A **Random Forest** algorithm is suitable because:

- Handles both numerical and categorical data.
- Provides high prediction accuracy.
- Reduces overfitting.
- Works well with large datasets.
- Is robust against noisy data.

Data Preprocessing

- Remove missing values.
- Remove duplicate records.
- Normalize numerical features.
- Encode categorical variables (weather, road type).
- Split the dataset into training and testing sets.

Feature Engineering

New useful features can be created such as:

- Peak hour indicator.
- Rain intensity.
- Average vehicle speed.
- Holiday/weekend indicator.

These features improve prediction performance.

Training and Validation

- Train the model using 80% of the data.
- Test using the remaining 20%.
- Apply cross-validation for reliable performance.
- Evaluate using Accuracy, Precision, Recall, and F1-score.

Deployment

The trained model is deployed on a cloud server where live traffic data from sensors and cameras is continuously processed. The system predicts congestion in real time and:

- Adjusts traffic signal timing.
- Suggests alternate routes.
- Sends alerts to navigation applications.

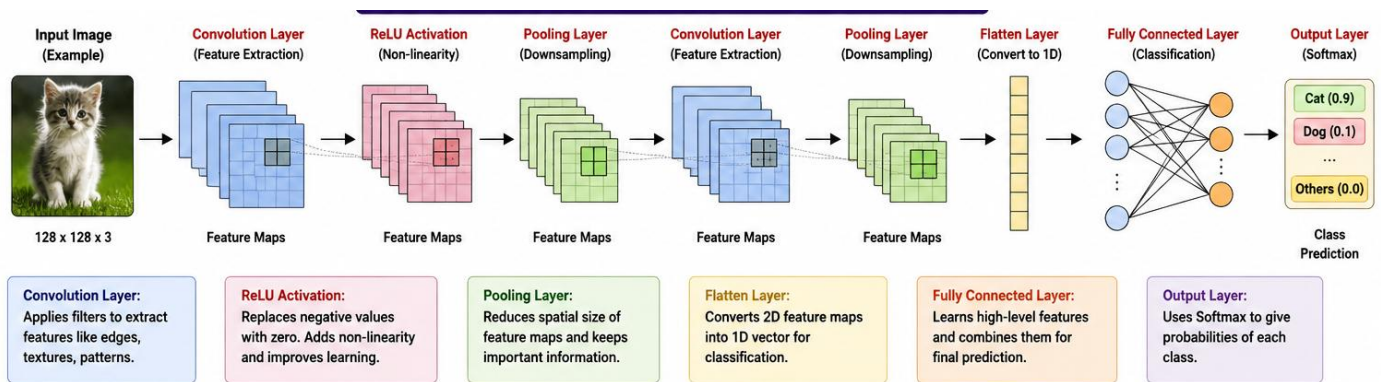
Conclusion

A machine learning-based smart traffic management system improves traffic flow, minimizes travel time, reduces fuel consumption, and enhances road safety through real-time congestion prediction.

3. **In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.**

Answer

A **Convolutional Neural Network (CNN)** is a deep learning model specifically designed for image recognition and computer vision tasks. It automatically extracts features from images and classifies them with high accuracy.



Layers in CNN

1. Convolution Layer

- Applies filters (kernels) to detect edges, corners, textures, and patterns.
- Produces feature maps.

2. ReLU Activation

- Replaces negative values with zero.
- Introduces non-linearity and speeds up learning.

3. Pooling Layer

- Reduces feature map dimensions.
- Lowers computational cost.
- Common methods: Max Pooling and Average Pooling.

4. Flatten Layer

- Converts the 2D feature maps into a 1D vector.

5. Fully Connected Layer

- Combines extracted features.
- Performs final classification using the Softmax function.

Feature Map Generation

Each convolution filter scans across the input image and calculates weighted sums. Different filters detect different image characteristics, producing multiple feature maps.

Backpropagation

During training:

1. The predicted output is compared with the actual label.
2. Loss is calculated.
3. Gradients are computed using backpropagation.
4. Weights are updated using Gradient Descent.
5. This process repeats until the model achieves minimum loss.

Image Classification Example

For a dataset containing **cats and dogs**:

- The CNN extracts low-level features like edges.
- Deeper layers identify ears, eyes, tails, and fur.
- The fully connected layer classifies the image as **Cat** or **Dog**.

Factors Affecting Performance

- Size and quality of the dataset.
- Number of convolution layers.
- Learning rate.
- Batch size.
- Number of training epochs.
- Data augmentation techniques.
- Regularization methods such as Dropout.

Conclusion

CNNs automatically learn image features without manual feature extraction. Their ability to learn hierarchical representations makes them highly effective for image classification, object detection, and medical image analysis.